

# Md Shaifur Rahman

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github.com/shaifurcodes | dev.to/shaifurcodes | plasso.ai | plasso.ai/portfolio

## SUMMARY

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Computer Science PhD with 10+ years building end-to-end embedded and wireless systems. Inventor on 4 US patents; 8 peer-reviewed publications (200+ citations, h-index 7). Expertise spans real-time firmware in C/C++ (ARM Cortex-M, STM32, FreeRTOS/Zephyr), sensor fusion (UWB/IMU/LoRa), AI inference on edge hardware (Jetson Orin Nano, llama.cpp/CUDA), and full-stack system integration from hardware bring-up to production deployment. Founder & CEO of Plasso, an AI persona robotics startup.

## TECHNICAL SKILLS

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|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Robotics</b>             | ROS 2, Gazebo, Nav2, OpenCV, SLAM, Sensor Fusion (LiDAR+IMU), Wheel Odometry.                                                               |
| <b>Sim2Real</b>             | MuJoCo playground, Lerobot, Isaac Lab                                                                                                       |
| <b>Embodied AI &amp; ML</b> | VLA, VLM & Edge LLM Deployment (Gemma, smolVLA on Jetpack), Imitation/Reinforcement Learning, Hugging Face Hub integration, Llama.cpp       |
| <b>Embedded Systems</b>     | C/C++, Python, Firmware, Low-level programming, Latency Optimization, wireless protocols, Linux Systems, SPI/I2C/Serial/CAN etc. interfaces |
| <b>Hardware &amp; Edge</b>  | NVIDIA Jetson Orin, 6-DOF Arms (SO-101), RPLIDARs, IMX Cameras, IMUs, 520 Motor Encoders, STS Servo motors                                  |
| <b>Prototype Design</b>     | FreeCAD, OpenSCAD, KiCAD                                                                                                                    |
| <b>Robotic CI/CD</b>        | Industrial_CI+GitHub-Action, Regression test, HIL-test                                                                                      |

## PROFESSIONAL EXPERIENCE

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### Plasso (plasso.ai)

*Founder & CEO*

Seattle, WA

*2025 – Present*

- Architecting an AI persona robot platform with swappable downloadable personalities running fully offline on NVIDIA Jetson Orin Nano; STT→LLM→TTS pipeline (Whisper.cpp → llama.cpp/Gemma → Piper) achieving <500 ms round-trip latency on-device.
- Built a custom CUDA 13.2-compatible llama.cpp from source for JetPack 7.2 (resolved container incompatibility); integrated DistilBERT zero-shot NLI classifier for real-time query routing in a Python AI agent, replacing fragile regex-based routing.
- Launched full company infrastructure: domain (plasso.ai), Netlify investor landing page, USPTO Class 009 trademark filing, GitHub repos, and social channels; targeting US Semiquincentennial (July 4, 2026) with a Lincoln Edition persona as launch wedge.
- Designed 4-DOF upper-body companion robot architecture with Qdrant-backed RAG and an App-Store-style persona marketplace; offline STT/LLM/TTS stack runs without cloud dependency.

### NEC Labs America

*Research Intern — Embedded Systems & Localization*

Princeton, NJ

*Jun 2019 – Apr 2021*

- Led firmware and algorithm development for NavigateIO, a zero-infrastructure 3D indoor tracking system for first responders, fusing UWB, IMU, LoRa, and Visual Odometry on ARM Cortex-M targets.
- Achieved sub-meter 3D localization accuracy in multi-story NLOS environments using COTS Qorvo DW1000 modules; drop-and-go anchor architecture enables deployment in under 60 seconds.
- Co-invented 4 granted US patents (12,335,899 · 12,207,226 · 11,595,934 · 11,388,564); contributed to securing \$1M in seed capital.
- Built complete embedded IoT stack: hardware bring-up, sensor drivers (SPI/I2C/UART), real-time data pipelines, and fault-tolerant diagnostics for field deployments.

### Stony Brook University — Wireless & Networks Lab

*Research Assistant (NSF-funded, 3 concurrent grants)*

Stony Brook, NY

*May 2014 – May 2019*

- **FSO-VR** (NSF #1815306, \$500K): Designed and prototyped a steerable Free Space Optics wireless link for AR/VR headsets using Galvo-mirror beam steering and real-time RSSI feedback; achieved 99.99% link availability at 8K VR bandwidth; led to a US patent.
- **FSONet** (NSF #1514017, \$720K): Built a 100 m FSO backhaul prototype with custom Tracking & Pointing mechanism; validated resilience against fog and vibration; formulated near-optimal transceiver placement algorithm for 3D

urban topologies (NYC, San Francisco); published at ACM MobiCom 2017.

- **SpecSense** (NSF #1642965, \$800K): Developed crowdsensed spectrum occupancy mapping using RTL-SDRs and spatial interpolation; designed temporal clustering for time-varying maps; published at IEEE INFOCOM 2017.

## Philips Research North America

Research Intern — IoT

Briarcliff, NY

Jun 2015 – Aug 2015

- Implemented IoT mesh network protocols optimized for 4G backhaul connectivity in a healthcare device context.

## Bangladesh University of Engineering and Technology

Lecturer, Computer Science

Dhaka, Bangladesh

May 2009 – Jun 2013

- Delivered courses on VLSI Design, AI, Theory of Computation, OS, Database Systems, and Microprocessor Design; served as CISCO CCNA 1–4 Trainer.
- Consulted for Central Bank of Bangladesh automation project and Bangladesh's first machine-readable passport system.

## EDUCATION

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### Stony Brook University

Ph.D., Computer Science

Stony Brook, NY

May 2025

- Dissertation: "License-free Use of RF and Infrared Spectrum" — Advisors: Prof. Himanshu Gupta & Prof. Samir Das
- Dept. Fellowship (2013–2014); ACM SIGCOMM, MobiSys & IEEE SMARTCOMP Travel Grant recipient

### Stony Brook University

M.S., Computer Science — GPA 3.78/4.0

Stony Brook, NY

Dec 2017

### Bangladesh University of Engineering and Technology

B.S., Computer Science — GPA 3.92/4.0 (Dean's List every semester)

Dhaka, Bangladesh

Mar 2009

## US PATENTS (ALL FROM NAVIGATEIO FIRST-RESPONDER TRACKING SYSTEM)

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- US 12,335,899: Simultaneous localization and synchronization across multiple antennas
- US 12,207,226: Device tracking with radio ranging and sensor functions
- US 11,595,934: Infrastructure-free tracking and response
- US 11,388,564: Infrastructure-free RF tracking in dynamic indoor environments

## SELECTED PUBLICATIONS (190+ CITATIONS, H-INDEX 7 — GOOGLE SCHOLAR)

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1. **Rahman** et al. "DynoLoc: Infrastructure-free RF Tracking in Dynamic Indoor Environments." *Under revision, IEEE Trans. Mobile Computing*, 2026.
2. **Rahman** et al. "FSONet: A Wireless Backhaul for Multi-Gigabit Picocells Using Steerable FSO." *ACM MobiCom*, 2017.
3. **Rahman** et al. "SpecSense: Crowdsensing for Efficient Querying of Spectrum Occupancy." *IEEE INFOCOM*, 2017.
4. **Rahman** et al. "FSO-VR: Steerable Free Space Optics Link for Virtual Reality Headsets." *ACM WearSys*, 2018.
5. Curran, **Rahman** et al. "Rethinking Virtual Network Embedding in Reconfigurable Networks." *IEEE SECON*, 2018.
6. Gupta, **Rahman**, Curran. "Spectrum Allocation via Pathloss Estimation in Crowdsensed Shared Spectrum Systems." *IEEE DySPAN*, 2019.
7. **Rahman** et al. "Creating Spatio-temporal Spectrum Maps from Sparse Crowdsensed Data." *IEEE WCNC*, 2019.
8. **Rahman**, Naznin. "Efficient Routing in a Sensor Network Using Collaborative Ants." *Springer ICSI*, 2016.